

Problem Reconstruction

The Problem (No Solution Named Yet)

AI assistants forget everything once a conversation ends. They do not remember what you told them before.

Example 1: Fixing code

A person shows the AI some code and asks for help fixing a bug. Later, the person asks another question about that same code. But the AI has forgotten the code completely, so the person has to paste it in again.

Example 2: Planning a trip

A person talks to the AI about a trip to Japan. They talk about which cities to visit and what dates to go. Later, in a new chat, the person asks, "What should I pack for my trip?" The AI does not remember the trip at all. It does not even know the person is going to Japan. The person has to explain everything again from the start.

Example 3: Mixing up two different people

A company gives all its employees access to one AI assistant. Employee A tells the AI a private fact, like their salary. Later, Employee B, a different person, asks the AI something unrelated. If the AI is not careful with its memory, it might accidentally bring up Employee A's private salary while talking to Employee B. This is dangerous. It breaks trust, and in some countries it could even break the law.

What goes wrong

- The person has to repeat things they already said before.
- This wastes the person's time.
- Sometimes the person forgets to repeat a small detail, and then the AI gives a wrong or incomplete answer.
- After this happens enough times, the person starts to trust the AI less. It feels broken, even though the AI is actually smart. The real problem is that it cannot remember. It is not that it cannot think.

- If the AI mixes up different people's information, it can leak private facts to the wrong person. That is a serious problem, not just an annoying one.
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What Limits Any Fix (Constraints)

A constraint is a rule the system can never break, no matter how good an idea sounds. These are not extra features. They are real-world limits that any fix has to work inside, the same way a house has to be built within the limits of its walls, its budget, and the weather.

1. The AI can't read unlimited text at once (token budget)

The AI can only look at a certain amount of text at one time. It cannot just keep every memory it has ever saved and read all of it every time someone asks a question.

Example: Imagine someone has talked to the AI every day for a year, about trips, about work, about a hundred other things. If the AI tried to read every single one of those old conversations before answering something as simple as "what should I pack?", it would not fit. There is a hard limit on how much text it can look at in one go, so it has to pick out only the memories that matter right now, not all of them.

2. People don't want to wait (latency)

Whatever the AI checks in its memory, it has to do quickly. If checking memory takes too long, the AI feels slow and annoying to use, even when the answer it eventually gives is correct.

Example: Imagine someone asks the AI to fix a bug, and the AI has to search through months of saved code and notes before it can reply. If that search takes thirty seconds every single time, the person gets frustrated waiting, even though the AI technically remembered the right thing. A slow memory can feel worse than no memory at all.

3. Every word costs money (cost)

Every bit of text the AI reads costs money to process. If the memory system keeps growing, and the AI has to check more and more saved information every time, it slowly becomes too expensive to run.

Example: Imagine a company gives every employee access to this AI. If the AI has to reread thousands of saved facts for every single question from every single employee, the company's bill for running it keeps climbing. This happens even when most of those saved facts are not relevant to the question being asked. The system has to be smart about only checking what it actually needs, or the cost spirals out of control.

4. Some things must stay private (privacy)

Some things a person shares should never be shown to anyone else. Not by accident, and not ever. If the AI remembers something private, it has to be very careful about who is allowed to see that memory later.

Example: Employee A tells the AI a private fact, like their salary. That memory belongs only to Employee A. If the memory system is not careful, it might bring that fact up while talking to Employee B, someone who was never supposed to know it. This is not just embarrassing. It can break trust, and in some places it can even break the law.

5. The AI shouldn't guess (correctness)

If the AI is not sure what it remembers, it should not just make something up and present it as a real memory. A wrong memory can be more harmful than admitting it does not remember at all.

Example: Imagine someone told the AI weeks ago, "I'm allergic to peanuts," but the memory got garbled or half forgotten over time. If the AI later guesses and confidently says "you're fine with peanuts," that is a dangerous wrong answer. It would be far better if the AI had simply said, "I don't have that saved, can you remind me?" A fuzzy, uncertain memory should never be presented as a fact.

6. Different people's information must stay separate (multi-user isolation)

If many different people use the same AI, one person's saved information must never mix into someone else's conversation. This is true even by accident, and even if their questions sound similar.

Example: If Employee A and Employee B both ask the AI similar-sounding questions, the AI has to be able to tell them apart and only bring back each person's own memories. If the system mixes them up because the questions "sound the same," one person's private information ends up in someone else's chat. This is the same danger as the privacy example above, but this time it is caused by the system failing to tell people apart, not by failing to protect a secret on purpose.